Runlong Yu

Tenure-Track Assistant Professor | Department of Computer Science
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Academic and Professional Appointments

Tenure-Track Assistant Professor — August 2025–present
Department of Computer Science, The University of Alabama — Tuscaloosa, USA

Postdoctoral Associate — September 2023–August 2025
Department of Computer Science, University of Pittsburgh — Pittsburgh, USA

Research Intern — July 2019–January 2020
Data Application Center, Technology Engineering Group, Tencent — Shenzhen, China

Education

Doctor of Engineering (Ph.D. equivalent), Computer Science and Technology — June 2023
University of Science and Technology of China (USTC) — Hefei, China

Bachelor of Engineering, Computer Science and Technology — June 2017
University of Science and Technology of China (USTC) — Hefei, China

Awarded Research Funding

NASA: Development of an End-User-Driven Inland Water Quality Monitoring and Decision-Support System Powered by NASA-IBM Prithvi-EO2 Foundation Model.

Co-Investigator; PI: Hongxing Liu; other Co-Is: Yuehan Lu and Christopher Impellitteri. Award No. 80NSSC26K0971. Total institutional award: $749,999. Award period: October 1, 2026–September 30, 2029.

Honors, Awards, and Distinctions

Best BlueSky Paper Award, First Prize, IEEE ICDM / Computing Community Consortium; "Truth Without Comprehension: A BlueSky Agenda for Steering the Fourth Mathematical Crisis" (with Xiaowei Jia). 2025

ALAAI Faculty Fellow for Frontier AI, Alabama Center for the Advancement of Artificial Intelligence. 2026

Best Paper Award Candidate, IEEE ICDM; "Learning to Retrieve for Environmental Knowledge Discovery." 2025

Gold Medal, Anhui Province "Internet+" College Student Innovation Competition. 2021

Champion, CCF BDCI Fighting Epidemics Big Data Charity Challenge. 2021

China National Scholarship, Ministry of Education of the People's Republic of China. 2019

PaddlePaddle Special Award, KDD Cup, Regular Machine Learning Track. 2019

Third Runner-up, China Big Data and Artificial Intelligence Innovation Competition. 2018

Suzhou Industrial Park Scholarship. 2018

GDC Technology Scholarship. 2018

First Prize, Chinese Physics Olympiad. 2012

SELECTED PUBLICATIONS

Runlong Yu is highlighted in author lists. * denotes equal contribution; † denotes corresponding author.

Publications Since Joining UA (2025–Present)

Journal Articles

1. **Runlong Yu**, Yiqun Xie, Xiaowei Jia. Environmental Computing as a Branch of Science, *Communications of the ACM (**CACM**)*, 68(7):92–94, 2025. doi:10.1145/3725982. *Impact Factor: 22.7*

Conference Articles

1. Yilong Dai, Yiming Sun, Yiheng Chen, Shengyu Chen, Peyman Givi, Xiaowei Jia, **Runlong Yu**†. TIDE: A Physically Diverse 3D Turbulence Benchmark Dataset for Advancing Scientific Machine Learning, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), Datasets and Benchmarks Track*, accepted, to be published, 2027. arXiv:2608.04222. *Cycle 1 Acceptance Rate: 22%*
2. Yilong Dai, Yiming Sun, Yiheng Chen, Ziyi Wang, Shengyu Chen, Xiaowei Jia, **Runlong Yu**†. Physics-Preserving Latent Compression for Zero-Shot Resolution Transfer in 3D Turbulence, *IEEE International Conference on Data Mining (**ICDM**), Research Track*, accepted, to be published, 2026. *Research Track Acceptance Rate: 19.9%*
3. Chonghao Qiu, Shiyuan Luo, Feng Zhu, **Runlong Yu**, Yiqun Xie, Xiaowei Jia. ISO-UNet: A Climate-State Conditioned Regional Expert Network for Water Isotope Field Regression, *IEEE International Conference on Data Mining (**ICDM**), Research Track*, accepted, to be published, 2026. *Research Track Acceptance Rate: 19.9%*
4. Yilong Dai, Shengyu Chen, Xiaowei Jia, **Runlong Yu**†. Flow Learners for PDEs: Toward a Physics-to-Physics Paradigm for Scientific Computing, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), Blue Sky Ideas Track*, pages 13164–13169, 2026. *Blue Sky Ideas Track Acceptance Rate: 23.6%*
5. Qi Cheng, Licheng Liu, Yixuan Chen, Qing Zhu, **Runlong Yu**, Zhenong Jin, Yiqun Xie, Xiaowei Jia. LLM-based Evaluation Policy Extraction for Ecological Modeling, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), AI4Sciences Track*, pp. 10704–10715, 2026. doi:10.1145/3770855.3819033. *AI4Sciences Track Acceptance Rate: 30.8%*
6. Yilong Dai, Shengyu Chen, Ziyi Wang, Xiaowei Jia, Yiqun Xie, Vipin Kumar, **Runlong Yu**†. Learning PDE Solvers with Physics and Data: A Unifying View of Physics-Informed Neural Networks and Neural Operators, *International Joint Conference on Artificial Intelligence (**IJCAI**), Survey Track*, pp. 7817–7825, 2026. doi:10.24963/ijcai.2026/868. *Survey Track Acceptance Rate: 20.9%*
7. Yiming Sun, Qi Cheng, Licheng Liu, **Runlong Yu**, Yiqun Xie, Xiaowei Jia. Retrieval-Augmented Multi-scale Framework for County-Level Crop Yield Prediction Across Large Regions, *SIAM International Conference on Data Mining (**SDM**)*, accepted, to be published, 2026. *Acceptance Rate: 26.0%*
8. Yiheng Chen, Zihui Ma, Peishi Jiang, Yilong Dai, Qikai Hu, Xinyue Ye, Lingyao Li, Rita Leal Sousa, **Runlong Yu**†. When Earth Foundation Models Meet Diffusion: An Application to Land Surface Temperature Super-Resolution, *SIAM International Conference on Data Mining (**SDM**)*, accepted, to be published, 2026. *Acceptance Rate: 26.0%*
9. Yilong Dai, Yiming Sun, Yiheng Chen, Shengyu Chen, Xiaowei Jia, **Runlong Yu**†. FlowRefiner: Flow Matching-Based Iterative Refinement for 3D Turbulent Flow Simulation, *SIAM International Conference on Data Mining (**SDM**)*, accepted, to be published, 2026. *Acceptance Rate: 26.0%*
10. Lingyao Li*†, **Runlong Yu***†, Qikai Hu, Bowei Li, Min Deng, Yang Zhou, Xiaowei Jia. From Pixels to Places: A Systematic Benchmark for Evaluating Image Geolocalization Ability in Large Language Models, *Conference on Empirical Methods in Natural Language Processing (**EMNLP**), Main Conference*, accepted, to be published, 2026. (*equal contribution) *Main Conference Acceptance Rate: 15.4%*

11. Yiheng Chen, Lingyao Li, Zihui Ma, Qikai Hu, Yilun Zhu, Min Deng, **Runlong Yu**†. Empowering LLM Agents with Geospatial Awareness: Toward Grounded Reasoning for Wildfire Response, *Conference on Empirical Methods in Natural Language Processing (**EMNLP**), Findings of the ACL*, accepted, to be published, 2026. *Overall Acceptance Rate (Main Conference + Findings): 29.7%*
12. Naiyi Li, Zihui Ma, **Runlong Yu**†, Lingyao Li†. LSDTs: LLM-Augmented Semantic Digital Twins for Adaptive Knowledge-Intensive Infrastructure Planning, *AAAI Conference on Artificial Intelligence (**AAAI**)*, pages 38871–38879, 2026. *AISI Track Acceptance Rate: 24.1%*
13. Shiyuan Luo, Chonghao Qiu, **Runlong Yu**, Yiqun Xie, Xiaowei Jia. GREAT: Generalizable Representation Enhancement via Auxiliary Transformations for Zero-Shot Environmental Prediction, *AAAI Conference on Artificial Intelligence (**AAAI**)*, pages 38998–39006, 2026. *AISI Track Acceptance Rate: 24.1%*
14. **Runlong Yu**, Xiaowei Jia. Truth Without Comprehension: A BlueSky Agenda for Steering the Fourth Mathematical Crisis, *IEEE International Conference on Data Mining Workshops (**ICDMW**), BlueSky Track*, pages 2472–2476, 2025. *Best BlueSky Paper Award (First Prize)*
15. Shiyuan Luo*, **Runlong Yu***, Chonghao Qiu, Rahul Ghosh, Robert Ladwig, Paul C. Hanson, Yiqun Xie, Xiaowei Jia. Learning to Retrieve for Environmental Knowledge Discovery: An Augmentation-Adaptive Self-Supervised Learning Framework, *IEEE International Conference on Data Mining (**ICDM**)*, pages 527–536, 2025. (*equal contribution) *Best Paper Award Candidate; Regular Paper Acceptance Rate: 13.5%*
16. Shiyuan Luo, **Runlong Yu**, Shengyu Chen, Yingda Fan, Yiqun Xie, Yanhua Li, Xiaowei Jia. Geo-Aware Models for Stream Temperature Prediction across Different Spatial Regions and Scales, *ACM International Conference on Advances in Geographic Information Systems (**ACM SIGSPATIAL**)*, pages 124–136, 2025. *Acceptance Rate: 26.0%*
17. Lingyao Li, Dawei Li, Zhenhui Ou, Xiaoran Xu, Jingxiao Liu, Zihui Ma, **Runlong Yu**, Min Deng. LLMs as World Models: Data-Driven and Human-Centered Pre-Event Simulation for Disaster Impact Assessment, *Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, pages 3078–3096, 2025. *Main Conference Acceptance Rate: 22.1%*

Publications Prior to Joining UA

Journal Articles

1. **Runlong Yu**, Qi Liu, Yuyang Ye, Mingyue Cheng, Enhong Chen, Jianhui Ma. Collaborative List-and-Pairwise Filtering from Implicit Feedback, *IEEE Transactions on Knowledge and Data Engineering (**IEEE TKDE**)*, v 34, n 6, pages 2667-2680, 2022. *Impact Factor: 8.9*
2. **Runlong Yu**, Hongke Zhao, Zhong Wang, Yuyang Ye, Peining Zhang, Qi Liu, Enhong Chen. Negatively Correlated Search with Asymmetry for Real-Parameter Optimization Problems, *Journal of Computer Research and Development (**J-CRAD**)*, v 56, n 8, pages 1746-1757, 2019. *Class A by China Computer Federation*
3. Yin Gu, Kai Zhang, Qi Liu, Haojie Yuan, **Runlong Yu**. π-eLight: Learning Interpretable Programmatic Policies for Effective Traffic Signal Control, *IEEE Transactions on Mobile Computing (**IEEE TMC**)*, v 25, n 1, pages 1122–1136, 2026. *Impact Factor: 7.7*
4. Xiaojie Li, **Runlong Yu**, Lei Chen, Shengjun Liu, Enhong Chen. Process Optimization of Multi-stage Continuous Production System Based on Feature Fusion Modeling, *IEEE Transactions on Automation Science and Engineering (**IEEE T-ASE**)*, v 22, pages 21512–21524, 2025. *Impact Factor: 5.4*
5. Kai Zhang, Hongbo Gang, Feng Hu, **Runlong Yu**, Qi Liu. Adaptive ensemble learning for efficient keyphrase extraction: Diagnosis, aggregation, and distillation, *Expert Systems with Applications (**ESWA**)*, v 278, Article 127236, 2025. *Impact Factor: 7.5*
6. Yuyang Ye, Hengshu Zhu, Tianyi Cui, **Runlong Yu**, Le Zhang, Hui Xiong. University Evaluation through Graduate Employment Prediction: An Influence based Graph Autoencoder Approach, *IEEE Transactions on Knowledge and Data Engineering (**IEEE TKDE**)*, v 36, n 11, pages 7255-7267, 2024. *Impact Factor: 8.9*
7. Yuyang Ye, Zheng Dong, Hengshu Zhu, Tong Xu, Xin Song, **Runlong Yu**, Hui Xiong. MANE: Organizational Network Embedding with Multiplex Attentive Neural Networks, *IEEE Transactions on Knowledge and Data Engineering (**IEEE TKDE**)*, v 35, n 4, pages 4047-4061, 2023. *Impact Factor: 8.9*

8. Huijie Liu, Han Wu, Le Zhang, **Runlong Yu**, Ye Liu, Chunli Liu, Minglei Li, Qi Liu, Enhong Chen. A Hierarchical Interactive Multi-channel Graph Neural Network for Technological Knowledge Flow Forecasting, *Knowledge and Information Systems (**KAIS**)*, v 64, n 7, pages 1723-1757, 2022. *Impact Factor: 2.5*
9. Qixiang Shao, **Runlong Yu**, Hongke Zhao, Chunli Liu, Mengyi Zhang, Hongmei Song, Qi Liu. Toward Intelligent Financial Advisors for Identifying Potential Clients: A Multitask Perspective, *Big Data Mining and Analytics (**BDMA**)*, v 5, n 1, pages 64-78, 2022. *Impact Factor: 11.8*

Conference Articles

1. **Runlong Yu***, Shengyu Chen*, Yiqun Xie, Xiaowei Jia. A Survey of Foundation Models for Environmental Science, *Pacific Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**)*, pages 39-57, 2025. (*equal contribution) *Accepted to Data Science: Foundations and Applications (DSFA) Session*
2. **Runlong Yu**, Yiqun Xie, Xiaowei Jia. What We Talk About When We Talk About AI for Science, *SIAM International Conference on Data Mining (**SDM**)*, pages 439-442, 2025. *Invited Blue Sky Idea*
3. **Runlong Yu**, Chonghao Qiu, Robert Ladwig, Paul C. Hanson, Yiqun Xie, Xiaowei Jia. Physics-Guided Foundation Model for Scientific Discovery: An Application to Aquatic Science, *AAAI Conference on Artificial Intelligence (**AAAI**)*, v 39, n 27, pages 28548-28556, 2025. *AISI Track Acceptance Rate: 23.3%*
4. **Runlong Yu**, Chonghao Qiu, Robert Ladwig, Paul C. Hanson, Yiqun Xie, Yanhua Li, Xiaowei Jia. Adaptive Process-Guided Learning: An Application in Predicting Lake DO Concentrations, *IEEE International Conference on Data Mining (**ICDM**)*, pages 580-589, 2024. *Regular Paper Acceptance Rate: 10.9%*
5. **Runlong Yu**, Robert Ladwig, Xiang Xu, Peijun Zhu, Paul C. Hanson, Yiqun Xie, Xiaowei Jia. Evolution-based Feature Selection for Predicting Dissolved Oxygen Concentrations in Lakes, *International Conference on Parallel Problem Solving from Nature (**PPSN**)*, pages 398-415, 2024. *Acceptance Rate: 34.4%*
6. **Runlong Yu**, Xiang Xu, Yuyang Ye, Qi Liu, Enhong Chen. Cognitive Evolutionary Search to Select Feature Interactions for Click-Through Rate Prediction, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**)*, pages 3151-3161, 2023. *Research Track Acceptance Rate: 22.1%*
7. **Runlong Yu**, Qi Liu, Yuyang Ye, Mingyue Cheng, Enhong Chen, Jianhui Ma. Collaborative List-and-Pairwise Filtering from Implicit Feedback (Extended Abstract), *IEEE International Conference on Data Engineering (**ICDE**)*, pages 3801-3802, 2023. *Invited Extended Abstract*
8. **Runlong Yu**, Yuyang Ye, Qi Liu, Zihan Wang, Chunfeng Yang, Yucheng Hu, Enhong Chen. XCrossNet: Feature Structure-Oriented Learning for Click-Through Rate Prediction, *Pacific Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**)*, pages 436-447, 2021. *Acceptance Rate: 20.4%*
9. **Runlong Yu**, Yunzhou Zhang, Yuyang Ye, Le Wu, Chao Wang, Qi Liu, Enhong Chen. Multiple Pairwise Ranking with Implicit Feedback, *ACM Conference on Information and Knowledge Management (**CIKM**)*, pages 1727-1730, 2018. *Short Paper Acceptance Rate: 23.4%*
10. Yiming Sun, **Runlong Yu**, Runxue Bao, Yiqun Xie, Ye Ye, Xiaowei Jia. Domain-Adaptive Continual Meta-Learning for Modeling Dynamical Systems: An Application in Environmental Ecosystems, *SIAM International Conference on Data Mining (**SDM**)*, pages 297-306, 2025. *Acceptance Rate: 26.7%*
11. Yingda Fan, **Runlong Yu**, Janet Rice Barclay, Alison P. Appling, Yiming Sun, Yiqun Xie, Xiaowei Jia. Multi-Scale Graph Learning for Anti-Sparse Downscaling, *AAAI Conference on Artificial Intelligence (**AAAI**)*, v 39, n 27, pages 27969-27977, 2025. *AISI Track Acceptance Rate: 23.3%*
12. Yuyang Ye, Zhi Zheng, Yishan Shen, Tianshu Wang, Hengruo Zhang, Peijun Zhu, **Runlong Yu**, Kai Zhang, Hui Xiong. Harnessing Multimodal Large Language Models for Multimodal Sequential Recommendation, *AAAI Conference on Artificial Intelligence (**AAAI**)*, v 39, n 12, pages 13069-13077, 2025. *Acceptance Rate: 23.4%*
13. Xiang Xu, Hao Wang, Wei Guo, Luankang Zhang, Wanshan Yang, **Runlong Yu**, Yong Liu, Defu Lian, Enhong Chen. Multi-granularity Interest Retrieval and Refinement Network for Long-Term User Behavior Modeling in CTR Prediction, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**)*, pages 2745-2755, 2025. *ADS Track Acceptance Rate: 22.0%*

14. Yin Gu, Kai Zhang, Qi Liu, **Runlong Yu**, Xin Lin, Xinjie Sun. ProCC: Programmatic Reinforcement Learning for Efficient and Transparent TCP Congestion Control, *ACM International Conference on Web Search and Data Mining (**WSDM**)*, pages 963-972, 2025. *Acceptance Rate: 23.4%*
15. Yuyang Ye, Lu-An Tang, Haoyu Wang, **Runlong Yu**, Wenchao Yu, Erhu He, Haifeng Chen, Hui Xiong. PAIL: Performance based Adversarial Imitation Learning Engine for Carbon Neutral Optimization, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**)*, pages 6148-6157, 2024. *ADS Track Acceptance Rate: 20.0%*
16. Yujie Chen, **Runlong Yu**, Qi Liu, Enhong Chen, Zhenya Huang. Using Entropy for Group Sampling in Pairwise Ranking from Implicit Feedback, *International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR**)*, pages 2496-2500, 2023. *Short Paper Acceptance Rate: 25.1%*
17. Xiaojie Li, **Runlong Yu**, Guiquan Liu, Lei Chen, Enhong Chen, Shengjun Liu. Research on Multi-objective Optimization Algorithm for Coal Blending, *China National Conference on Big Data and Social Computing (**BDSC**)*, pages 37-60, 2023. *Regular Paper Acceptance Rate: 16.3%*
18. Yang Yu, Qi Liu, Likang Wu, **Runlong Yu**, Lei Yu, Zaixi Zhang. Untargeted Attack against Federated Recommendation Systems via Poisonous Item Embeddings and the Defense, *AAAI Conference on Artificial Intelligence (**AAAI**)*, pages 4854-4863, 2023. *Acceptance Rate: 19.6%*
19. Zhikang Mo, Qixiang Shao, Likang Wu, **Runlong Yu**, Jiexin Xu, Hongmei Song, Enhong Chen. TGNN: A GNN-Based Method with Multi-entity Node for Personal Banking Time Prediction, *Intelligent Networked Things: 5th China Conference (**CINT**)*, pages 68-79, 2023. *Acceptance Rate: 34.6%*
20. Yuren Zhang, Enhong Chen, Binbin Jin, Hao Wang, Min Hou, Wei Huang, **Runlong Yu**. Clustering based Behavior Sampling with Long Sequential Data for CTR Prediction, *International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR**)*, pages 2195-2200, 2022. *Short Paper Acceptance Rate: 24.7%*
21. Zheng Gong, Shiwei Tong, Han Wu, Qi Liu, Hanqing Tao, Wei Huang, **Runlong Yu**. Tipster: A Topic-Guided Language Model for Topic-Aware Text Segmentation, *International Conference on Database Systems for Advanced Applications (**DASFAA**)*, pages 213-221, 2022. *Acceptance Rate: 18.0%*
22. Huijie Liu, Han Wu, Le Zhang, **Runlong Yu**, Ye Liu, Chunli Liu, Qi Liu, Enhong Chen. Technological Knowledge Flow Forecasting through A Hierarchical Interactive Graph Neural Network, *IEEE International Conference on Data Mining (**ICDM**)*, pages 389-398, 2021. *Regular Paper Acceptance Rate: 9.9%*
23. Mingyue Cheng, Fajie Yuan, Qi Liu, Shenyang Ge, Zhi Li, **Runlong Yu**, Defu Lian, Senchao Yuan, Enhong Chen. Learning Recommender Systems with Implicit Feedback via Soft Target Enhancement, *International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR**)*, pages 575-584, 2021. *Regular Paper Acceptance Rate: 21.0%*
24. Shiwei Tong, Qi Liu, **Runlong Yu**, Wei Huang, Zhenya Huang, Zachary A Pardos, Weijie Jiang. Item Response Ranking for Cognitive Diagnosis, *International Joint Conference on Artificial Intelligence (**IJCAI**)*, pages 1750-1756, 2021. *Acceptance Rate: 13.9%*
25. Yuyang Ye, Hengshu Zhu, Tong Xu, Fuzhen Zhuang, **Runlong Yu**, Hui Xiong. Identifying High Potential Talent: A Neural Network based Dynamic Social Profiling Approach, *IEEE International Conference on Data Mining (**ICDM**)*, pages 718-727, 2019. *Regular Paper Acceptance Rate: 9.1%*
26. Mingyue Cheng, **Runlong Yu**, Qi Liu, Vincent W. Zheng, Hongke Zhao, Hefu Zhang, Enhong Chen. Alpha-Beta Sampling for Pairwise Ranking in One-Class Collaborative Filtering, *IEEE International Conference on Data Mining (**ICDM**)*, pages 1000-1005, 2019. *Short Paper Acceptance Rate: 18.5%*
27. Han Wu, Kun Zhang, Guangyi Lv, **Runlong Yu**, Weihao Zhao, Enhong Chen, Jianhui Ma. Deep Technology Tracing for High-tech Companies, *IEEE International Conference on Data Mining (**ICDM**)*, pages 1396-1401, 2019. *Short Paper Acceptance Rate: 18.5%*

Books & Chapters

1. Tong Xu, **Runlong Yu**. Epidemic Prevention and Control Big Data Cloud Platform: Solutions for Major Public Health Emergencies, *Annual Report on Development of Big Data Applications in China No.4 (2020) (**Blue Book of Big Data Applications**)*, Social Sciences Academic Press, ISBN 978-7-5201-7651-4, 2020.

RESEARCH PROPOSALS AND GRANT DEVELOPMENT

Amounts below are total requested project budgets. Project periods are proposed and contingent on funding.

Submitted Full Proposals – Pending Decisions

[P1] **AIDatasets:Impact: Building the Data Foundation for Reliable Neural Surrogates of Three-Dimensional Turbulence.** National Science Foundation. **Lead PI**; co-PIs: Jeffrey C. Carver (UA), Xiaowei Jia (Rutgers), Peyman Givi and Masoud Barati (Pittsburgh). Requested: $1,888,080; proposed period: September 2027–August 2030. Submitted November 4, 2026; NSF proposal 2705027.

[P2] **AIDatasets:Impact: FLUXNET-EB: Transforming the Global Evapotranspiration Flux Archive into an Energy-Balance-Corrected, AI-Ready Community Dataset.** National Science Foundation. **Co-PI**; PI: Mukesh Kumar. Requested: $1,765,610; proposed period: September 2027–August 2030. Submitted November 4, 2026; NSF proposal 2705265.

[P3] **AIDatasets:Impact: WhoGoesWhere: Integrating Mobility and Census Data for AI-Enabled Scientific Discovery.** National Science Foundation. **Investigator**; PI: Xiaowei Chen. Requested: $1,996,536; proposed period: June 2027–May 2030. Submitted November 4, 2026; NSF proposal 2704997.

[P4] **Learning to Generate Video and Sound from Shared Physical Causes.** Amazon Research Awards, Fall 2026. **PI**; co-PI: Kaushik Roy. Requested: $80,000 in research funding and $40,000 in AWS promotional credits. Submitted November 1, 2026; proposal P-2026-5957.

Earlier Submitted Full Proposals

- **Cross-Scale Prediction and Control of Energy–Water Systems Under Extreme Events: The Oroville Dam Case**, *U.S. Department of Energy (DOE)*, $997,922, 2026–2029. **PI.**
- **User-Centered Housing Risk Decision Support Using Prithvi Earth Foundation Models**, *National Aeronautics and Space Administration (NASA)*, $494,754, 2026–2028. **PI; Co-PIs: T. Loof and P. Jiang.**
- **Efficient Agentic AI for Sustainable Forestry, Land Intelligence, and Mill Operations**, *National Science Foundation (NSF)*, $1,749,999, 2026–2028. **Co-PI; PI: Jackey Gong; Co-PI: J. Geyer.**
- **AI Companion Robot Research: User, Intelligence, and Form Development**, *University of South Carolina*, $68,653, 2026–2028. **PI.**
- **UTC – THRIVE: Tunnel Health, Resilience, Inspection, & Viability Extension**, *Colorado School of Mines*, $300,001, 2026–2027. **Co-PI; PI: Xinyue Ye.**
- **A Paradigm Shift for Water Quality Prediction: Building Ontology and Knowledge Graphs**, *CIROH / National Oceanic and Atmospheric Administration (NOAA)*, $60,000. **Co-PI; PI: X. Hong; Co-PI: T. N. Le.**

Proposals Not Selected for Funding

- **Collaborative Research: CAIG: From Synthetic Cities to Real Cities: Physics-Guided Multi-Fidelity Reconstruction of Extreme Heatwave Evolution**, *National Science Foundation (NSF)*, $592,041, 2026–2029. **PI; Co-PI: Xinyue Ye.**
- **Collaborative Research: CAIG: Dual Threats of Droughts and Heatwaves: A Physics-Structured Graph Operator for Riverine Hydro-Thermal Processes**, *National Science Foundation (NSF)*, $593,783, 2026–2029. **Co-PI; PI: Peishi Jiang.**
- **Collaborative Research: FIRE-WUI: Enabling Immersive Experiences of WUI Wildfire Risks to Accelerate House Adaptation**, *National Science Foundation (NSF)*, $217,845, 2026–2028. **PI.**
- **Topic 16-A Phase I: Transferable Transmission-Grid Foundation Models for Scalable Contingency Screening and Infrastructure Planning**, *U.S. Department of Energy (DOE)*, $750,000, 2026–2027. **PI; Co-PIs: B. Balasubramanian, L. Wang, and S. Li.**

- **Topic 16-B Phase I: Physics-Certified Agentic Decision Support for Near-Real-Time Control of Active Distribution Systems**, *U.S. Department of Energy (DOE)*, $750,001, 2026–2027. **Co-PI; PI: L. Wang; Co-PIs: S. Li and B. Balasubramanian.**

WHITE PAPERS

[W1] **Improving Pesticide Concentration Prediction in Sparsely Monitored Watersheds Using Data Companion Hydrology. PI**. Submitted October 13, 2026.

[W2] **Prioritizing Water Quality Monitoring for Veterinary Antibiotics in Agricultural Watersheds. PI**. Submitted October 13, 2026.

[W3] **FIMsens-Agent: Autonomous Remote Sensing Flood Inundation Mapping with Large Earth Foundation Models and AI Agents for Rapid Disaster Response. Co-Investigator**; PI: Dan Tian. Submitted October 14, 2026.

[W4] **Developing and Evaluating Computer Vision and Deep-learning Tools for Water Monitoring with CIR and TIR Cameras in USGS HIF and Field Environments. Co-Investigator**; PI: Hongxing Liu. Submitted October 14, 2026.

[W5] **Operational Safety Assurance of CIROH in the Era of Generative AI. Co-Investigator**; PI: Jiaqi Gong. Submitted October 14, 2026.

[W6] **Low-Cost Edge AI-Enabled Smart Gage for Rapid-Deployment Flood Monitoring of Small Streams. Co-Investigator**; PI: Na Gong. Submitted October 13, 2026.

[W7] **AI-Ready Coastal Testbeds: Connecting NOAA Water Predictions to Ecosystem Data Products. Co-Investigator**; PI: Xinyue Ye. Submitted October 13, 2026.

TEACHING EXPERIENCE

Artificial Intelligence — Fall 2026 (ongoing)
Instructor; undergraduate, graduate, and online courses — The University of Alabama

Physics-Guided Machine Learning — Spring 2026
Instructor; graduate course — The University of Alabama

Principles of Data Mining — Spring 2025
Instructor; graduate course — University of Pittsburgh

Introduction to Data Science — Fall 2024
Instructor; undergraduate course — University of Pittsburgh

Principles of Data Mining — Spring 2024
Guest teacher; graduate course — University of Pittsburgh

Machine Learning and Data Mining — Fall 2022; Spring 2023
Teaching assistant; graduate course — USTC

Computer Science and Society — Fall 2020; Spring 2021
Teaching assistant; undergraduate course — USTC

Database Systems and Applications — Spring 2017
Teaching assistant; undergraduate course — USTC

RESEARCH MENTORING

The University of Alabama. Graduate mentees: Yilong Dai, Yiheng Chen, Tanvir Ahammad, and Udbhav Srivastava. Undergraduate mentee: Riley Herchert.

University of Pittsburgh. Graduate mentees: Shiyuan Luo, Chonghao Qiu, Xiaoting Li, Yingda Fan, and Yiming Sun. Undergraduate mentee: Mariia Sierova.

University of Science and Technology of China. Graduate mentees: Qixiang Shao, Yujie Chen, Xiang Xu, Xiaojie Li, Zhikang Mo, Yang Yu, and Yuren Zhang.

SELECTED TALKS AND CONFERENCE PRESENTATIONS

- "AI for Scientific Discovery in an Open World: Learning Structure in Complex Natural Systems." Invited Limnology Seminar, Center for Limnology, University of Wisconsin–Madison, Madison, WI, October 7, 2026.
- "Advance Water Monitoring Using Machine Learning." Environmental Data Initiative (EDI) Power Users Meeting, 2026.
- "Geospatial Intelligence in Large Language Models." OM–ST Ph.D. Seminar, Department of Information Systems, Statistics, and Management Science, The University of Alabama, 2026.
- "Truth Without Comprehension: A BlueSky Agenda for Steering the Fourth Mathematical Crisis." IEEE ICDM, 2025.
- "What We Talk About When We Talk About AI for Science." SIAM SDM, 2025.
- "Adaptive Process-Guided Learning: An Application in Predicting Lake DO Concentrations." KGML Workshop, 2024.
- "Cognitive Evolutionary Search to Select Feature Interactions for Click-Through Rate Prediction." ACM SIGKDD, 2023.
- "XCrossNet: Feature Structure-Oriented Learning for Click-Through Rate Prediction." PAKDD, 2021.
- "Negatively Correlated Search with Asymmetry for Real-Parameter Optimization Problems." CCFAI, 2019.

PEER REVIEW AND PROFESSIONAL SERVICE

Conference Program Committee and Reviewer Service. AAAI: 2021–2026; IJCAI: 2021–2026; ICML: 2022–2026. NeurIPS: 2022–2024, 2026; ICLR: 2024–2025. KDD: 2026 (AI4Sciences Track); ACL Rolling Review: 2026. The Web Conference (WWW): 2025–2026; PAKDD: 2025–2026; AISTATS: 2025–2026.

Journal Peer Review. IEEE Transactions on Pattern Analysis and Machine Intelligence; IEEE Transactions on Knowledge and Data Engineering; IEEE Transactions on Big Data; IEEE Transactions on Systems, Man, and Cybernetics: Systems; ACM Computing Surveys; ACM Transactions on Knowledge Discovery from Data; ACM Transactions on Recommender Systems; Communications of the ACM; Advanced Science.

Invitations for the 2027 Conference Cycle. AAAI 2027 (Main Conference and AI for Social Impact Track); KDD 2027 (AI4Sciences and Datasets and Benchmarks Tracks); IJCAI 2027 (Main Track).

University Service. Member, Computer Science Curriculum Committee (undergraduate curriculum), The University of Alabama, August 2025–September 2026.

Member, Special Committee: Space & Infrastructure, Department of Computer Science, The University of Alabama, 2026–2027.